# A Deep Learning Framework for Identifying Essential Proteins Based on Vision Transformer

Yuqing Mao, Gaoshi Li, Xu Lin, Jingli Wu, Wei Peng, Jiafei Liu, and Xiaoshu Zhu

***Abstract*—Essential proteins are fundamental to the reproduction and survival of cells, and if they are killed, the cells will stop reproducing or die. Many computational methods of identifying essential proteins are proposed which fuse a large number of features from multi-omics data. Some of them extract features from subcellular localization data by subjectively selecting certain subcellular locations. Meanwhile, there is still room to improve the identification rate of essential proteins. In this paper, a new deep learning framework for identifying essential proteins based on Vision Transformer is proposed, named EPViT. Firstly, topological features are extracted from the protein-protein interaction network. Secondly, a feature matrix is designed from the subcellular localization information without subjective factor. Then, the two classes of features are fused into a new feature matrix by outer product operation. Finally, the new feature matrix is input into the Vision Transformer model to discover essential proteins. The results show that EPViT has the highest recognition rate among the comparison experiments on yeast data.**

***Index Terms*—Deep learning framework, essential proteins, vision transformer, protein-protein interaction, subcellular localization.**

## I. Introduction

ESSENTIAL proteins are fundamental to the reproduction and survival of cells, and if they are killed, the cells will stop reproducing or die [1]. Identifying essential proteins is important in revealing cellular molecular mechanisms [2] and discovering novel drug targets [3]. There are several methods of identifying essential proteins have been proposed. Generally, the identification methods can be divided into biological experimental methods and computational methods.

In the past several decades, biological experimental methods, including gene knockouts [4], RNA interference [5], and conditional knockouts [6], have identified a number of essential proteins. But time-consuming, low-efficiency, and high-cost are their common shortcomings.

Computational techniques overcome these shortcomings with biological experimental methods. Researchers extract features from biological data produced by high-throughput technologies. For single feature-based methods, a serial centrality methods [7], [8], [9], [10], [11], [12], [13] are extracted from protein-protein interaction (PPI) data. However, they have low discovery capacity. For traditional methods, researchers find that fusing multiple features may obtain a higher identification rate. Some new features are abstracted from multi-omics biological information, including gene expression profiles [14], [15], subcellular localization information [16], [17], orthologous information [18], [19], protein complex information [20], and Go annotation [21]. These features are widely used to combine with PPI data to identify essential proteins. Although the proposed traditional methods [22], [23], [24], [25], [26] have higher identification rates than single feature-based methods, their fusion models limit the number of features, and their discovery rate is still low.

Machine learning-based methods have the advantage of fusing a large number of features. Support vector machine (SVM) [27], decision tree [28], and ensemble learning [29] are widely used. Yet, it is necessary to subjectively select some features from a large number of existing biological features, then these features are input into certain machine learning models. Consequently, deep learning-based methods including [16], [30], [31] are applied on predicting essential proteins. They are commonly end-to-end methods, and the work of extracting features is usually done automatically. Although these deep learning-based methods have good performance in identifying essential proteins, there is still room to improve the recognition rate.

Many methods [16], [31] use subcellular localization information. The subcellular localization information of a protein are related to cellular function and other possible interactions with proteins [32]. For feature representation of subcellular localization information, Zeng et al. [16] subjectively select the common 11 subcellular locations as the feature spaces. And only 0 and 1 are used to encode each protein into an 11-dimensional vector.

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Even though Yue et al. [31] select more subcellular locations (1 024), they sort the subcellular locations in descending order according to the number of proteins for each subcellular location. Those methods extract features from subcellular localization data by subjectively selecting certain subcellular locations.

While feature extraction is a well-established paradigm in bioinformatics, the manner of fusing multi-omics data remains a critical design choice. Recent studies underline this point: Liu et al. [33] demonstrate that enhancing embedding distinguishability in a latent space improves peptide identification. Bai et al. [34] observe that conventional fusion strategies—such as concatenation or attention—often neglect high-order interactions between feature subspaces in protein function prediction. The graph-based approaches relying on simple feature merging [35] lack explicit mechanisms to model structured cross-feature dependencies. Therefore, the manner of the features fusion is one of the important factors that affect the recognition results.

The Vision Transformer (ViT) [36] model has been applied in other recognition fields. ViT has advantages over other deep learning models in dealing with imbalanced binary classification problems: 1) Global context modeling via self-attention. Unlike convolutional neural networks (CNNs), which rely on local receptive fields and may overlook long-range dependencies, ViT employs self-attention mechanisms that allow each token to attend to all other tokens globally. This is particularly important in biological networks such as PPI networks, where essential proteins often exert influence across distant nodes. For example, a hub protein might interact with functionally diverse partners located far apart in the network topology—such global relationships can be effectively captured by ViT. 2) Compatibility with high-dimensional feature matrices. These feature matrices represented proteins can be flattened into sequences of patches before being fed into ViT. The patch-based design of ViT naturally accommodates high-dimensional input representations, making it suitable for handling complex fused embeddings derived from biological data. Traditional models like CNNs or MLPs would struggle to process such structured matrices efficiently without significant architectural modifications. These advantages enable ViT to perform better for identifying essential proteins.

To obtain a higher identification rate, a new deep learning framework for identifying essential proteins based on ViT, named EPViT, is proposed in this paper. In this framework, the biological data consist of PPI data and subcellular localization data. EPViT has these steps: 1) fusing features by outer product operation, which includes three processes in this part: i) extracting topological features from PPI data by using node2vec technique [37]; ii) generating an indicator vector for subcellular localization data without subjective factor; iii) merging the above features by outer product operation; 2) training the ViT model and identifying essential proteins. The results suggest that EPViT shows better performance than other existing methods on yeast data.

## II. Materials and Methods

In this section, the data materials and methods with EPViT are introduced. The data resources are detailed in Section II-A. Then the overview of EPViT is illustrated in Section II-B. More details of the overview are explained in Sections II-C, II-D, and II-E. Finally, the evaluation metrics of the experiments are listed in Section II-F.

### A. Data Materials

Three PPI datasets, a subcellular localization dataset, and an essential proteins golden standard dataset of Saccharomyces cerevisiae (S.cerevisiae) are used in our work. The first PPI dataset is downloaded from DIP database [38]. There are 5093 proteins and 24743 interactions after filtering self-interactions and repeated interactions. From the work of Krogan et al. [39], there are two PPI datasets. There are 3672 and 2708 proteins, and 14317 and 7123 interactions in the two datasets, respectively. The three PPI datasets are named as PPI_5093, PPI_3672, and PPI_2708 in our study, respectively.

Subcellular localization dataset of yeast is derived from the integrated channels of COMPARTMENTS database [40] that records confidence level scores of 737045 proteins appearing at each subcellular localization. Through counting, there are 2702 subcellular locations in this dataset.

The essential proteins golden standard dataset is fused from four databases: MIPS [41], SGD [42], DEG [2], and SGDP [43]. It has 1285 essential proteins. Specifically, 1167, 928, and 785 essential proteins are contained in PPI_5093, PPI_3672, and PPI_2708, respectively.

### B. Overview Of Epvit

Fig. 1 illustrates the overview of EPViT. There are two types of biological data: PPI and subcellular localization, which are the inputs to EPViT. The first step in EPViT is to extract features from the two types of biological information separately. The node2vec technique is employed to automatically extract topological features from PPI network. For subcellular localization information, each protein is encoded as an indicator vector with contiguous values. After extracting the two types of features, the outer product operation is utilized to combine them into a feature matrix. Then, the feature matrix is input to train the ViT model, and all proteins are classified as essential or non-essential.

### C. Product Embedding

In order to fuse two types of features from PPI data and subcellular localization data, we design a way called product embedding in this paper. The process of product embedding is introduced in detail as follows.

*1) Features Extraction From PPI Data:* The node2vec technique [37] is used to automatically extract topological features from PPI data. Node2vec is a graph embedding method that integrates Depth First Search (DFS) neighborhoods and Breadth First Search (BFS) neighborhoods and maps nodes in a complex network into a low-dimensional feature space. It captures neighborhoods and structural information between proteins from PPI data by performing a $2^{nd}$ order random walk strategy.

Firstly, the PPI data is denoted as network $G = (V, E)$, where $V$ and $E$ are the sets of vertexes (proteins) and edges (interactions), respectively. In this process, the context (neighbor node) associated with the protein is taken into account to generate

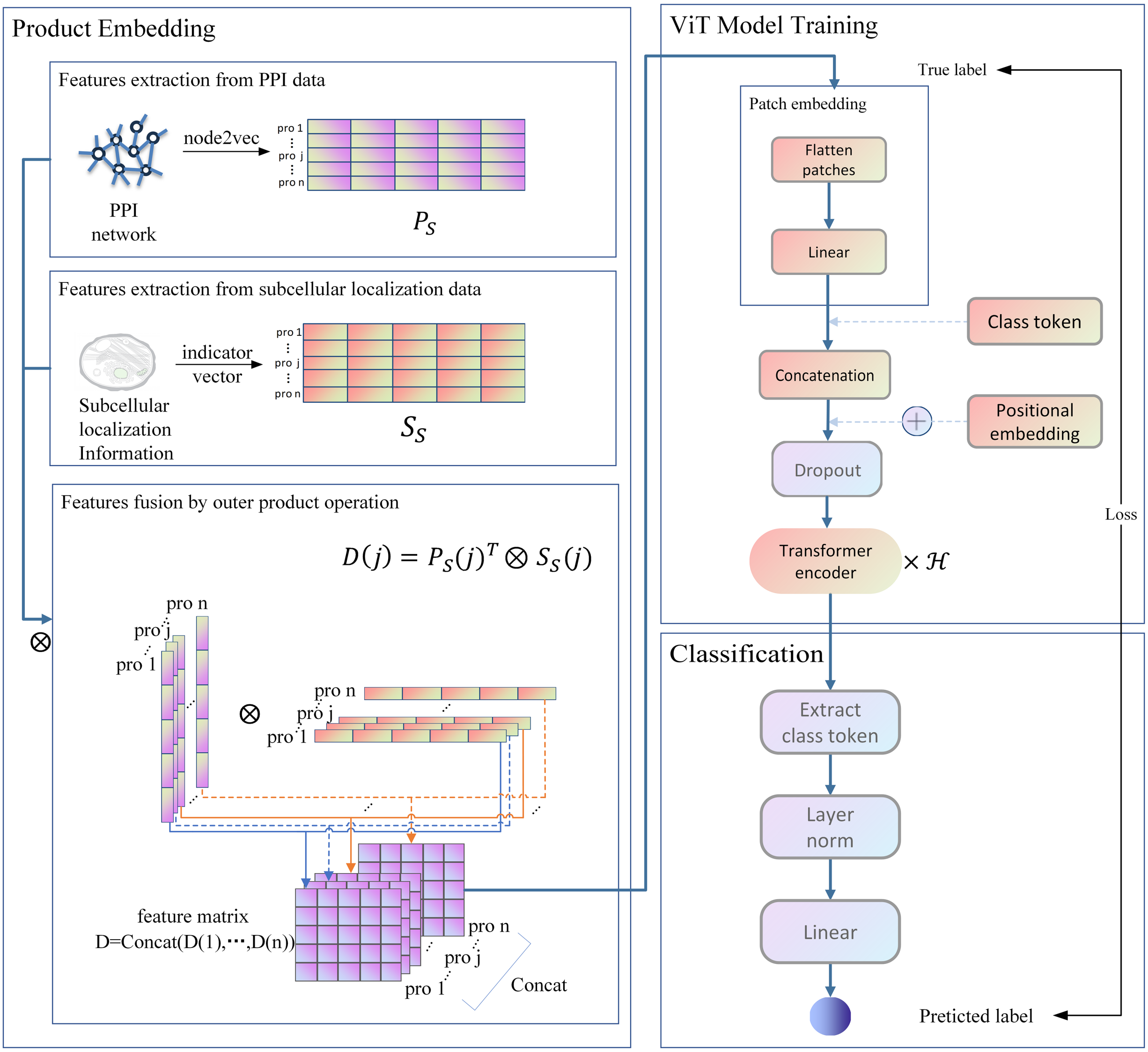



Fig. 1. The overview of EPViT. In features extraction from PPI data, the node2vec technique is used to automatically extract topological features from PPI data. The matrix $P_S$ represents the feature matrix extracted from PPI data using the node2vec technique, which is then normalized. In features extraction from subcellular localization data. The matrix $S_S$ indicates the feature matrix extracted from subcellular localization data, which is then normalized. In features fusion by outer product operation, $P_S(j)^T$ refers to the transposed embedding vector of protein $j$ in PPI data obtained by node2vec. $S_S(j)$ refers to the indicator vector extracted of protein $j$ in subcellular localization data. The character of $\otimes$ indicates the outer product operation. $D(j)$ is defined as being a new feature matrix representation of protein $j$ by fusing matrix $P_S$ and matrix $S_S$. The abbreviation pro refers to protein. In ViT model training, the character $\oplus$ indicates the addition operation. $\mathcal{H}$ is the number of transformer block.

a $d$-dimensional vector representation for each protein. The feature dimension $d$ is set as 224 in this study (same setting for $d$ mentioned in the following context). The probability of node $x$ to be sampled into the walk sequence follows the distribution:

$$P\left\{c_i = x \left| c_{i-1} = v\right.\right\} = \begin{cases} \frac{\pi_{vx}}{Z}, & \text{if } (\mathrm{v},\mathrm{x}) \in \mathrm{E}; \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

where $\pi_{vx}$ denotes the unnormalized transition probability between node $v$ and node $x$, $Z$ is a constant number used to normalize, and $c_i$ denotes the $i$th node in the walk sequence. Researchers invoke return parameter $p$ and in-out parameter $q$ to control the search procedure bias between BFS and DFS. The unnormalized transition probability is set to:

$$\pi_{vx} = \alpha_{pq}\ (t, x) \cdot \omega_{vx} \tag{2}$$

$$\text{where } \alpha_{pq}\ \{t, x\} = \begin{cases} \frac{1}{p}, & \text{if } d_{tx} = 0; \\ 1, & \text{if } d_{tx} = 1; \\ \{\frac{1}{q}, & \text{if } d_{tx} = 2. \end{cases} \tag{3}$$

And $d_{tx}$ denotes the shortest path distance between node $t$ and node $x$, and $\omega_{vx}$ is the edge weight.

After extracting the features from PPI data, a matrix $P \in \mathbb{R}^{n \times d}$ is obtained, which has $n$ rows and $d$ columns, where row denotes the sample (protein) and column is the feature space. And the matrix $P$ is normalized to matrix $P_S$ using the following

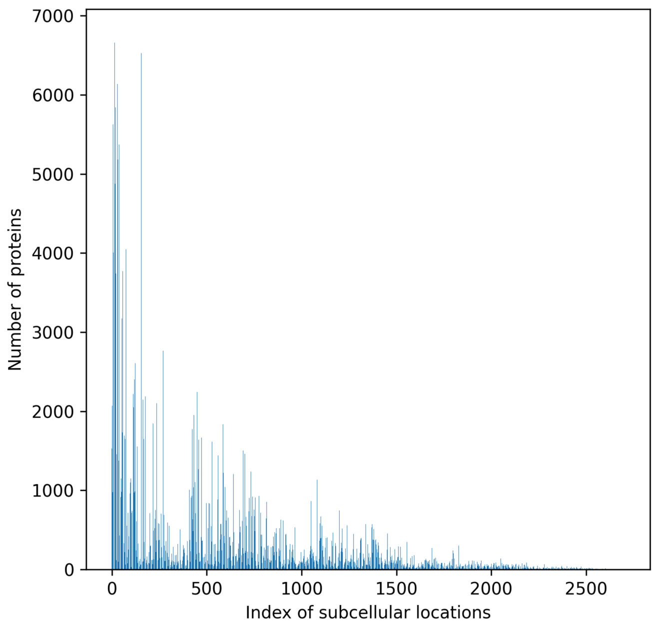



Fig. 2. The distribution of protein amounts in subcellular locations. Due to the excessive length of subcellular location names, the term index is used as a replacement for subcellular location, with the x-axis representing the index of the subcellular location. The y-axis represents the number of proteins at that subcellular location.

equation:

$$P_S\ (j,r) = \frac{P\,(j,r) - \min\left(P\,(j)\right)}{\max\left(P\,(j)\right) - \min\left(P\,(j)\right)}\ , j \in [1,n]\,,\ r \in [1,d] \tag{4}$$

where $P_S(j,r)$ indicates the normalized value of row $j$ and column $r$ in matrix $P_S$, $P(j) \in \mathbb{R}^{1\times d}$ refers to the embedding vector of protein $j$ in PPI data obtained by node2vec, $\min()$ is the minimum value function, $\max()$ is the maximum value function.

*2) Features Extraction From Subcellular Localization Data:* Subcellular localization information refers to the localization of proteins within cells. Different subcellular structures and locations provide distinct functional environments and interaction mechanisms. Interacted proteins must be co-localized in specific subcellular locations. By understanding the subcellular localization information of proteins, researchers can infer their cellular environments and involvement in biological processes.

The subcellular localization data contains a large number of subcellular locations (>2 700). The distribution of proteins appearing at each subcellular location in the subcellular localization data is illustrated in Fig. 2, and the x-axis is index of subcellular locations. It is clear that the subcellular locations with a high number of proteins are concentrated in the top 10% in this subcellular localization data.

Fig. 3 demonstrates the process of extracting features from subcellular localization data. In order to generate the new feature matrix which size is $d \times d$, the top $d$ subcellular locations are selected as the feature space. In Fig. 3, firstly, the subcellular localization data matrix $S \in \mathbb{R}^{M\times 1}$ is normalized to matrix $\bar{S}_S \in \mathbb{R}^{M\times 1}$ by (5), where $M$ refers to the number of proteins

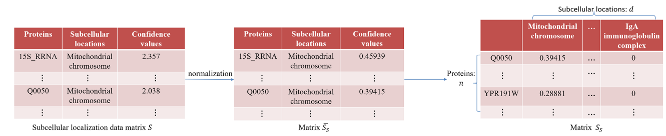



Fig. 3. Features extraction from subcellular localization data. $d$ denotes the dimension of matrix $S_S$.

in original file.

$$\bar{S}_S\ (k) = \frac{S\,(k) - \min\,(S)}{\max\,(S) - \min\,(S)}\ ,\ k \in [1, M] \tag{5}$$

In (5), $k$ denotes the protein contained in the matrix $S$. Then, the confidence level values of proteins in PPI data at the top $d$ subcellular locations are selected from matrix $\bar{S}_S$ and encoded for each protein with a $d$-dimensional vector. After extracting the features from subcellular localization data, a matrix $S_S \in \mathbb{R}^{n\times d}$ is obtained, which has $n$ rows and $d$ columns, where row denotes the sample (protein) and column is the feature space. One can see, these subcellular locations in our method are selected without subjective factor.

*3) Features Fusion:* The features from PPI data and subcellular localization data are integrated by applying outer product operation for each protein. Here is the equation of features combination, $D(j)$ is defined as being a new feature matrix representation of protein $j$ by fusing matrix $P_S$ and matrix $S_S$:

$$D\ (j) =\ P_S(j)^T \otimes S_S\,(j)\,,\ j \in V \tag{6}$$

where $P_S(j)^T$ refers to the transposed embedding vector of protein $j$ in PPI data obtained by node2vec, $S_S(j)$ refers to the indicator vector extracted of protein $j$ in subcellular localization data, the character of $\otimes$ in (6) indicates the outer product operation. The new feature matrix $D(j)$ provides a richer feature representation due to fuse two kinds of biological features. $D(j) \in \mathbb{R}^{H\times W\times C}$ $(H = W = d, C = 1)$ can be regarded as an image that is fed into the following ViT model, where $(H, W)$ is the size of image (feature matrix), $C$ is the number of channels. Like black-and-white image, the channel of the image is set to be a constant number 1. The feature matrices of $n$ proteins are concatenated as D $= (D(1), \cdots, D(n))$, where $n$ is the number of proteins in PPI data.

Unlike most graph-based or sequence-based methods that concatenate or add features, EPViT introduces an outer product operation to fuse topological (from PPI via node2vec) and subcellular localization features. This generates a 224×224 matrix for each protein that explicitly encodes pairwise interactions between dimensions of the two feature spaces—effectively creating a "feature interaction map." EPViT utilizes a structured, image-like representation—generated by fusing heterogeneous biological priors through an outer product—as input to a ViT for protein analysis.

## D. Vit Model

ViT model [36] comes from Transformer [44], which is mainly used in natural language processing based on the attention mechanism. ViT model has excellent performance on image classification task. In this paper, the feature matrix that combines PPI data and subcellular localization data is input into the ViT model as an image.

Fig. 1 shows the process of ViT Model Training. ViT model transforms a computer vision problem into a seq2seq problem through patch embedding. The image (feature matrix) $D(j) \in \mathbb{R}^{H\times W\times C}$ is reshaped into a sequence of flattened patches $D(j)_p \in \mathbb{R}^{N\times(P^2\cdot C)}$, where $P$ is the patch size with the value of 16, $N = HW/P^2$ is the number of patches. The patches of an image equal to the words of a sentence. Consequently, concatenating class token $D(j)_{class}$ with patches $D(j)_p = (D(j)_p^1, \ldots, D(j)_p^N)$ and adding positional embedding matrix $E_{pos}$ by position. Details of implementation is shown as (7), the new feature matrix of protein $j$ is represented as $z_j \in \mathbb{R}^{(N+1)\times dim}$:

$$z_j = \left[D(j)_{class}; D\,(j)_p^1\, E; D\,(j)_p^2\, E; \ldots; D\,(j)_p^N\, E\right] + E_{pos} \tag{7}$$

where $D(j)_{class} \in \mathbb{R}^{1\times dim}$, $E \in \mathbb{R}^{(P^2\cdot C)\times dim}$ is the linear transformation matrix, $E_{pos} \in \mathbb{R}^{(N+1)\times dim}$ is the learnable embedding matrix that concludes the positional information, and $dim$ is the dimension of the model, which is set as 1024 in this research. Then, the representations $z = (z_1, \ldots, z_n)$ of $n$ proteins are processed by the Transformer block. In Fig. 1, the Transformer block is stacked $\mathcal{H}$ times, which is set as 8 in this paper.

The Transformer block in ViT model is a major contributor and based solely on attention mechanism. (8) is the definition of the scaled dot-product attention function, or self-attention function:

$$Attention\ (Q, K, V) = \ softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{8}$$

where the matrix $Q$ consists a set of queries, matrices $K$ and $V$ are also pack keys and values separately, $d_k$ denotes the dimension of keys, which is set as 64. $Q$ and $K$ are calculated by the dot-product operation to obtain a similarity matrix, which is then scaled using $1/\sqrt{d_k}$. Afterwards, applying the softmax operation on the scaled similarity matrix to be as the weight matrix. Finally, the weight matrix is used to compute dot-product with $V$.

Based on the self-attention, multi-head attention contains more subspaces information. Each head represents a subspace. The multi-head attention function is defined as below:

$$MultiHead\ (Q, K, V) = \ Concat\,(head_1, \cdots, head_h)\, W^O \tag{9}$$

where $head_i = Attention(QW_i^Q,\ KW_i^K,\ VW_i^V),\ i \in (1, \cdots, h)$

where $Q, K, V \in \mathbb{R}^{(N+1)\times hd_k}$, the projections are learnable parameter matrices $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{hd_k\times d_k}$, $and\ W^O \in \mathbb{R}^{hd_v\times dim}$, $h$ is the number of heads, which is set as 12, $d_v$ denotes the dimension of values, which is set as 64. Fig. 4 is the illustration of multi-head attention.

After the model is trained, the class token is extracted and fed into the classification block to predict the result.

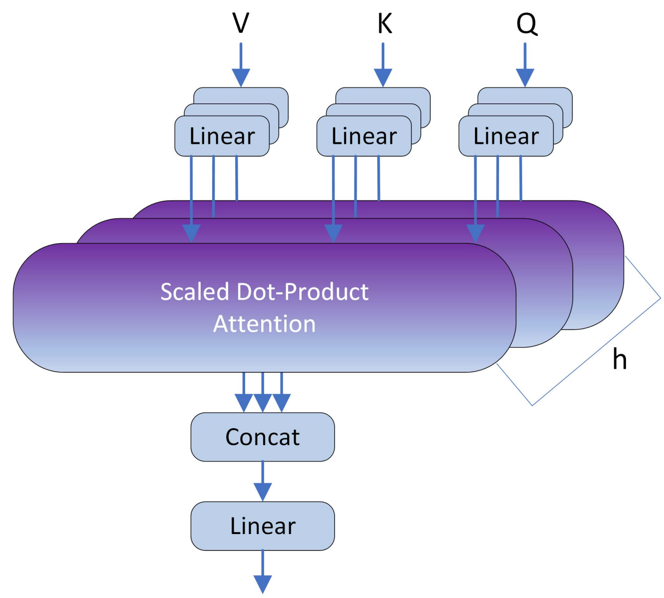



Fig. 4. Multi-head attention. $h$ is the number of heads.

### E. Loss Function

Imbalanced dataset tends to affect performance of the model. When the distribution of samples is imbalanced, the distribution of the loss function will be inclined, e.g., when $m \ll u$,where $m$ and $u$ denote the number of positive samples (essential proteins) and negative samples (non-essential proteins), respectively, negative samples will dominate the loss function. Due to the inclined loss function, the model training process will prefer the categories with negative samples, resulting in poorer performance of the model for positive samples. To tackle this problem, the focal loss function [45] is employed to calculate the loss while training. Here is the definition of focal loss function:

$$FL\ (\hat{y}, y) = \begin{cases} -\alpha(1-\hat{y})^{\gamma}\log(\hat{y}), & \text{if } y = 1; \\ -\left(1-\alpha\right)\hat{y}^{\gamma}\log(1-\hat{y}), & \text{otherwise.} \end{cases} \tag{10}$$

where $\hat{y}$ and $y$ denote the predicted label and true label, respectively, $\alpha$ is the balanced factor used to increase the weight of positive samples in the loss function and balance the distribution of the loss function. Focusing parameter $\gamma$ concerns easy-hard samples. The value of the parameter $\gamma$ is used as recommended in the original paper and is set to 2 on all PPI datasets. The value of parameter $\alpha$ is determined by the ratio of negative and positive samples:

$$\frac{\alpha}{1-\alpha} = \frac{u}{m} \tag{11}$$

where $u$ is the number of negative samples, $m$ is the number of positive samples. In the experiments, we set $\alpha$as 0.77, 0.74, and 0.71 on PPI_5093, PPI_3672, and PPI_2708, respectively.

### F. Performance Evaluation Metrics

In evaluating the performance of algorithms of predicting essential proteins, some metrics are commonly used, such as Accuracy, Precision, Recall, F-measure. Their definitions are defined as follows:

$$Accuracy = \frac{TN + TP}{TN + TP + FP + FN} \tag{12}$$

$$Precision = \frac{TP}{TP + FP} \tag{13}$$

$$Recall = \frac{TP}{TP + FN} \tag{14}$$

$$F - measure = \frac{\left(1 + \beta^2\right) \cdot Precision \cdot Recall}{\beta^2 \cdot Precision + Recall} \tag{15}$$

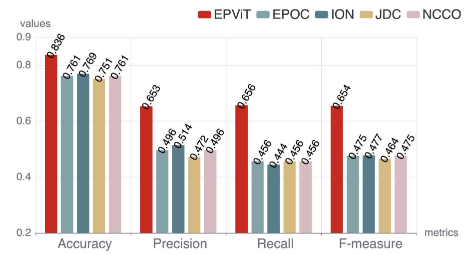
where *TP* (true positive) denotes the number of positive samples (essential proteins) predicted as positive. *TN* (true negative) is the number of negative samples (non-essential proteins) predicted as negative. *FP* (false positive) denotes the number of negative samples predicted as positive. *FN* (false negative) is the number of positive samples predicted as negative. In (15), $\beta$ ($\beta > 0$) is a parameter to adjust the weight between Recall and Precision. We set $\beta = 1$ in our experiment, which means that Recall and Precision are equally important.

Meanwhile, the Receiver Operating Characteristic (ROC) [46] curve and Precision-Recall (PR) [47] curve are also evaluation tools for binary classification that visualize the performance of the model. AUROC is the area under the ROC curve, AUPRC is the area under the PR curve. The models with larger AUROC and AUPRC values have better performance.

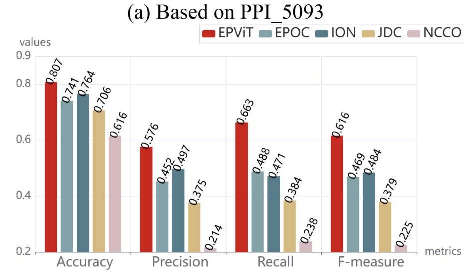
## III Results

In this section, the experiment results of EPViT are compared with seven methods for predicting essential proteins on the three PPI datasets. The versions of PyTorch framework and Python are 2.0.1 and 3.8.16, respectively. The optimizer is selected as Adam. The optimal hyperparameters combination is as follows: learning rate is 0.00005, weight decay is 0.0001, dropout rate is 0.2, epoch is 100, batch size is 32, random seed is 42. More specifically, the dataset is divided into 80% training set and 20% test set. The training set is used to train the model by applying $K$-fold cross-validation (each fold maintains the same proportion of essential and non-essential proteins as the full dataset to avoid overfitting to abnormally distributed subsets), where $K$ is set as 10. During training, early stopping (patience = 100) is implemented to terminate training if validation performance plateaus, and this measure is to prevent overfitting. The separate test set is used to evaluate the performance of the model. Meanwhile, the final performance results are averaged 10 times by 10-fold cross-validation. The values of AUROC and AUPRC for each fold on three PPI datasets are presented in Supplementary Materials Table S1-S3.

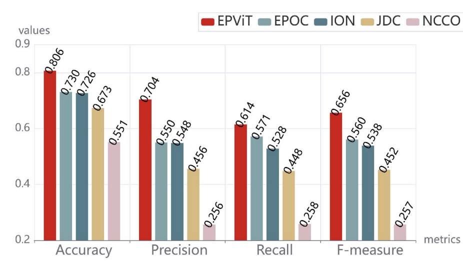
Sections III-A and III-B provide comparisons of the four traditional methods and the three deep learning-based methods, respectively. Then, the ablation studies on different multi-omics data and different loss functions are analyzed in Section III-C. It should be explained that different from deep-learning based methods, traditional methods do not divide the training set and test set. The traditional methods score all proteins and then sort them in descending order. The top $m$ (same as the $m$ mentioned in the last paragraph of Section II-E) proteins are predicted as essential proteins, and the remaining proteins are predicted as non-essential proteins.




(a) Based on PPI_5093




(b) Based on PPI_3672



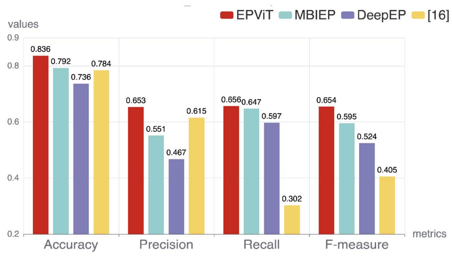

(c) Based on PPI_2708

Fig. 5. Comparison EPViT with other traditional methods on three PPI datasets. EPViT performs best compared to these multi-feature fusion methods. (a) The experiment results of EPViT, EPOC, ION, JDC, and NCCO on PPI_5093 dataset. (b) The experiment results of EPViT, EPOC, ION, JDC, and NCCO on PPI_3672 dataset. (c) The experiment results of EPViT, EPOC, ION, JDC, and NCCO on PPI_2708 dataset.

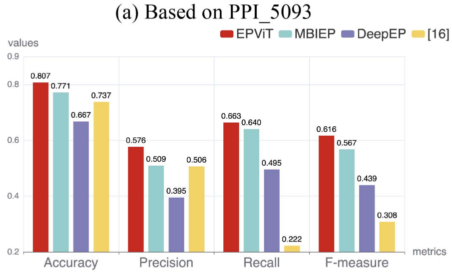
### A. Comparison With Traditional Methods

Fig. 5 intuitively illustrates the comparison with these methods (EPOC [22], ION [23], JDC [24], and NCCO [25]) based on three PPI datasets. In Fig. 5(a), the values of Accuracy, Precision, Recall, and F-measure of EPViT on PPI_5093 are 0.836, 0.653, 0.656, and 0.654, which are better than that of EPOC (0.761, 0.496, 0.456, and 0.475), respectively. Compared with these values of ION (0.769, 0.514, 0.444, and 0.477), JDC (0.751, 0.472, 0.456, and 0.464), and NCCO (0.761, 0.496, 0.456, and 0.475), EPViT improves by 8.7%, 27%, 47.8%, and 37.1%; by 11.3%, 38.4%, 43.9%, and 41%; by 9.9%, 31.7%, 43.9%, and 37.7%, respectively.

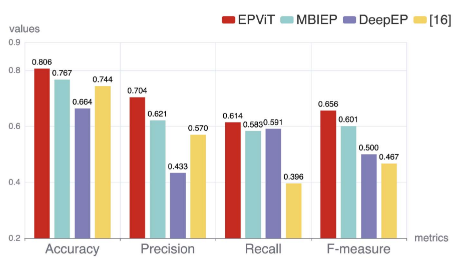
Fig. 5(b) shows the Accuracy, Precision, Recall, and F-measure values of EPViT on PPI_3672 are 0.807, 0.576, 0.663, and 0.616, respectively. Compared with the values of EPOC (0.741, 0.452, 0.488, and 0.469), ION (0.764, 0.497, 0.471, and




(a) Based on PPI_5093

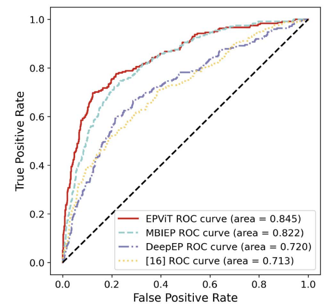



(b) Based on PPI_3672

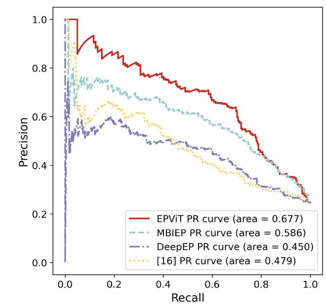



(c) Based on PPI_2708

Fig. 6. Comparison EPViT with other deep learning-based methods on three PPI datasets. EPViT performs best compared to these deep learning-based methods. (a) The experiment results of MBIEP, DeepEP, and [16] on PPI_5093 dataset. (b) The experiment results of MBIEP, DeepEP, and [16] on PPI_3672 dataset. (c) The experiment results of MBIEP, DeepEP, and [16] on PPI_2708 dataset.

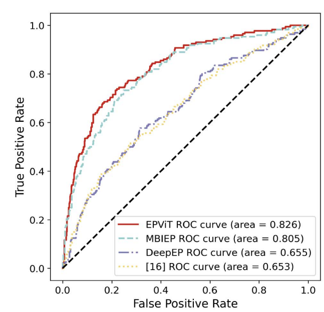



(a)ROC curves

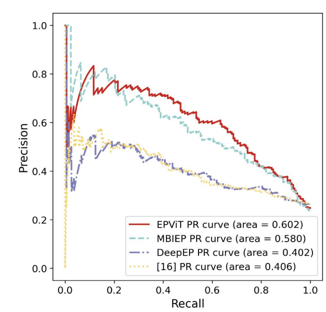



(b)PR curves

Fig. 7. ROC and PR curves of deep learning-based methods on PPI_5093. (a) It is the ROC curves of EPViT and MBIEP, DeepEP, and [16] on PPI_5093 dataset. The x-axis denotes the false positive rate. The y-axis denotes the true positive rate. (b) It is the PR curves of EPViT and MBIEP, DeepEP, and [16] on PPI_5093 dataset. The x-axis denotes the recall. The y-axis denotes the precision.




(a)ROC curves




(b)PR curves

Fig. 8. ROC and PR curves of deep learning-based methods on PPI_3672. (a) It is the ROC curves of EPViT and MBIEP, DeepEP, and [16] on PPI_3672 dataset. The x-axis denotes the false positive rate. The y-axis denotes the true positive rate. (b) It is the PR curves of EPViT and MBIEP, DeepEP, and [16] on PPI_3672 dataset. The x-axis denotes the recall. The y-axis denotes the precision.

0.484), JDC (0.706, 0.375, 0.384, and 0.379), EPViT has obvious advantage over these methods in all metrics. Furthermore, the advantage of EPViT is more obvious compared to NCCO (0.616, 0.214, 0.238, and 0.225), which improves by 31%, 169.2%, 178.6%, and 173.8%, respectively.

Like Fig. 5(a) and (b), (c) shows EPViT keeps the advantage in all metrics on PPI_2708. It shows the values of EPViT are 0.806, 0.704, 0.614, and 0.656, respectively; that of EPOC are 0.730, 0.550, 0.571, and 0.560, respectively; that of ION are 0.726, 0.548, 0.528, and 0.538, respectively; that of JDC are 0.673, 0.456, 0.448, and 0.452, respectively; that of NCCO are 0.551, 0.256, 0.258, and 0.257, respectively.

Overall, EPViT performs best compared to these multi-feature fusion methods on three PPI datasets.

## B. Comparison With Deep Learning-based Methods

Fig. 6 shows the results of comparison on deep learning-based methods (MBIEP [31], DeepEP [30], [16], and EPViT) based on three PPI datasets, respectively. In Fig. 6(a), the values of Accuracy, Precision, Recall, and F-measure of EPViT on PPI_5093 are 0.836, 0.653, 0.656, and 0.654, respectively; while that of MBIEP are 0.792, 0.551, 0.647, and 0.595, respectively. And EPViT improves by 13.6%, 39.8%, 9.9%, and 24.8% than that of DeepEP (0.736, 0.467, 0.597, and 0.524), respectively; by 6.6%, 6.2%, 117.2%, and 61.5% than that of [16] (0.784, 0.615, 0.302, and 0.405), respectively.

In Fig. 6(b), the values of EPViT on PPI_3672 are 0.807, 0.576, 0.663, and 0.616, which are better than that of MBIEP (0.771, 0.509, 0.640, and 0.567), DeepEP (0.667, 0.395, 0.495, and 0.439), and [16] (0.737, 0.506, 0.222, and 0.308).

It is easy to see EPViT has obvious advantage over other methods on PPI_2708 in Fig. 6(c). The values of EPViT are 0.806, 0.704, 0.614, and 0.656, respectively. While that of MBIEP are 0.767, 0.621, 0.583, and 0.601; that of DeepEP are 0.664, 0.433, 0.591, and 0.500; that of [16] are 0.744, 0.570, 0.396, and 0.467, respectively.

The figures (Figs. 7– 9) illustrate the ROC and PR curves of EPViT and other deep learning-based methods across three PPI datasets, respectively, where the AUROC and AUPRC values for each method are provided. It is obvious that the AUROC and AUPR values of EPViT are higher than that of other deep learning-based methods on three PPI datasets, respectively.

TABLE I
COMPARISON OF FOCAL LOSS FUNCTION AND CROSS-ENTROPY LOSS FUNCTION ON THREE PPI DATASETS

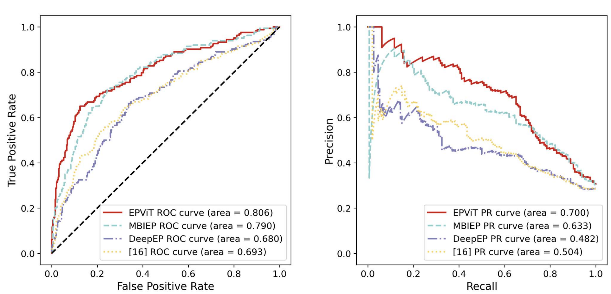
| Datasets | Loss functions | Accuracy | Precision | Recall | F-measure | AUROC | AUPRC |
|---|---|---|---|---|---|---|---|
| PPI_5093 | Focal loss | **0.836** | 0.653 | **0.656** | **0.654** | **0.845** | **0.677** |
| | Cross-entropy loss | 0.828 | **0.683** | 0.510 | 0.584 | 0.828 | 0.653 |
| PPI_3672 | Focal loss | 0.807 | 0.576 | **0.663** | **0.616** | **0.826** | **0.602** |
| | Cross-entropy loss | **0.813** | **0.606** | 0.581 | 0.594 | 0.820 | 0.596 |
| PPI_2708 | Focal loss | **0.806** | 0.704 | **0.614** | **0.656** | **0.806** | **0.700** |
| | Cross-entropy loss | 0.791 | **0.708** | 0.522 | 0.601 | 0.784 | 0.682 |

Bold values indicate the best performance.

TABLE II
COMPARISON OF DIFFERENT MULTI-OMICS DATA ON THREE PPI DATASETS

| Datasets | Multi-omics data | Accuracy | Precision | Recall | F-measure | AUROC | AUPRC |
|---|---|---|---|---|---|---|---|
| PPI_5093 | PPI+subcellular | **0.836** | **0.653** | 0.656 | **0.654** | **0.845** | **0.677** |
| | Only PPI | 0.237 | 0.237 | **1** | 0.383 | 0.587 | 0.319 |
| | Only subcellular | 0.819 | 0.611 | 0.652 | 0.631 | 0.838 | 0.659 |
| PPI_3672 | PPI+subcellular | **0.807** | **0.576** | 0.663 | **0.616** | **0.826** | **0.602** |
| | Only PPI | 0.349 | 0.237 | **0.802** | 0.366 | 0.513 | 0.256 |
| | Only subcellular | 0.781 | 0.527 | 0.634 | 0.575 | 0.819 | 0.580 |
| PPI_2708 | PPI+subcellular | **0.806** | **0.704** | 0.614 | **0.656** | 0.806 | 0.700 |
| | Only PPI | 0.301 | 0.301 | **1** | 0.463 | 0.551 | 0.375 |
| | Only subcellular | 0.778 | 0.637 | 0.614 | 0.625 | **0.833** | **0.722** |

Bold values indicate the best performance.




(a) ROC curves (b) PR curves

Fig. 9. ROC and PR curves of deep learning-based methods on PPI_2708. (a) It is the ROC curves of EPViT and MBIEP, DeepEP, and [16] on PPI_2708 dataset. The x-axis denotes the false positive rate. The y-axis denotes the true positive rate. (b) It is the PR curves of EPViT and MBIEP, DeepEP, and [16] on PPI_2708 dataset. The x-axis denotes the recall. The y-axis denotes the precision.

## C. Ablation Studies

To select a better loss function, the influences of focal loss function and cross-entropy loss function in this model are analyzed in Section III-C-1. To evaluate that the combination of PPI data and subcellular localization data enhances the performance of the model, ablation experiments are analyzed on three PPI datasets based on different multi-omics data, and the results are presented in Section III-C-2.

*1) Ablation Experiments On Different Loss Functions:* In Table I, based on PPI_5093, the values of Accuracy, Precision, Recall, F-measure, AUROC, and AUPRC with focal loss function are 0.836, 0.653, 0.656, 0.654, 0.845 and 0.677, which is higher than cross-entropy loss function (0.828, 0.683, 0.510, 0.584, 0.828, and 0.653), the Precision value of focal loss function is slightly lower than that of cross-entropy loss function. On PPI_3672, the values of Recall and F-measure with focal loss function are 0.663 and 0.616, compared with that of cross-entropy loss function (0.581 and 0.594), they improve by 14.1% and 3.7%, respectively. On PPI_2708, the values of Accuracy, Recall, F-measure, AUROC and AUPRC with focal loss function (0.806, 0.614, 0.656, 0.806 and 0.700) are higher than that of cross-entropy loss function (0.791, 0.522, 0.601, 0.784, and 0.682), the Precision value with focal loss function (0.704) is slightly lower than that of cross-entropy loss function (0.708). It clearly shows that the performance of focal loss function is better than cross-entropy loss function in this model.

*2) Ablation Experiments On Different Multi-Omics Data:* In Table II, on the PPI_5093 and PPI_3672, the combination of PPI data and subcellular localization data performs best on the Accuracy, Precision, F-measure, AUROC, and AUPRC metrics, which are 0.836, 0.653, 0.654, 0.845, and 0.677 on PPI_5093, and 0.807, 0.576, 0.616, 0.826, 0.602 on PPI_3672. This is better than both only PPI data and only subcellular localization data.

With only PPI data, the metrics of Accuracy, Precision, F-measure, AUROC, and AUPRC on PPI_5093 are 0.237, 0.237, 0.383, 0.587, and 0.319, on PPI_3672 are 0.349, 0.237, 0.366, 0.513, and 0.256, respectively.

In the case of only using subcellular localization data, Accuracy, Precision, F-measure, AUROC, and AUPRC have values of 0.819, 0.611, 0.631, 0.838, and 0.659 on the PPI_5093 and 0.781, 0.527, 0.575, 0.819, and 0.580 on the PPI_3672, respectively.

On the PPI_2708, the combination of PPI data and subcellular localization data has the best results for the Accuracy, Precision, and F-measure metrics, which are 0.806, 0.704, and 0.656, respectively. However, in the case where only subcellular localization data is used, the AUROC and AUPRC metric values are 0.833 and 0.722, respectively, which are higher than the results of the combination of PPI data and subcellular localization data (0.806 and 0.700, respectively).

Interestingly, the results show that using only the PPI data does not yield good results. It has high Recall values in all three PPI datasets, to even 1, indicating that predicting all

TABLE III
HYPERPARAMETER ANALYSIS ON PPI_5093 DATASET

| Hyperparameter | Value | AUC | AUPR |
|---|---|---|---|
| learning rate | 0.00001 | 0.830 | 0.659 |
| | 0.00005 | **0.845** | **0.677** |
| | 0.0001 | 0.825 | 0.655 |
| weight decay | 0.00001 | **0.845** | 0.672 |
| | 0.0001 | **0.845** | **0.677** |
| | 0.001 | 0.787 | 0.551 |
| dropout rate | 0.1 | 0.835 | 0.652 |
| | 0.2 | **0.845** | **0.677** |
| | 0.3 | 0.841 | 0.672 |

Bold values indicate the best performance.

samples as positive samples, and does not conduct the distinction between essential and non-essential proteins well. The results on PPI_2708 when using only subcellular localization data, although not as good as EPViT on PPI_5093 and PPI_3672, may indicating that subcellular localization data contributes more than PPI data when using the outer product operation for feature fusion in EPViT.

In general, the combination of PPI data and subcellular localization data has indeed improved the performance of the model.

*3) Hyperparameter Analysis:* The hyperparameter analysis focuses on three core hyperparameters that directly impact model performance: learning rate, weight decay, and dropout rate. All experiments are performed on the PPI_5093 dataset using AUC and AUPR as the evaluation metrics, with other parameters held constant in single-parameter analysis. And the results are summarized in Table III. As shown, the optimal configuration (learning rate = 0.00005, weight decay = 0.0001, dropout rate = 0.2) consistently achieves the highest AUC (0.806) and AUPR (0.700) across all tested parameters.

## IV Conclusion And Discussions

Identifying essential proteins is important in developing new antibiotics and prompting synthetic biology. In recent years, researchers have proposed a variety of computational approaches to identify essential proteins, while their identification rates still have room for improvement. In this paper, we propose a new deep learning framework for identifying essential proteins based on ViT model, named EPViT. The comparison experiments are done on yeast data, and the results show that 1) EPViT has the best identification rate than other comparison methods; 2) the performance of EPViT is stable on multiple PPI datasets; 3) EPViT has stable results on each running. EPViT's novelty is not merely the adoption of ViT, but the co-design of a biologically grounded input representation (via outer product) and a vision-inspired architecture to solve a classical systems biology problem in a new paradigm. This enables more expressive modeling of protein essentiality than either conventional network analysis or standard Transformer applications. However, EPViT needs slightly more hyperparameters and training time.

Moreover, there are several aspects of EPViT that can be improved in the future: 1) Incorporate additional multi-omics data into the model, such as orthologous information, gene expression profiles, etc. Currently, EPViT can only process one or two types of multi-omics data, and both types data provide limited diversity for the fused data features. On this basis, we will attempt to develop a new fusion model to integrate more multi-omics data; 2) The subcellular localization data will be investigated in more depth in our future work. The results of the ablation experiments using only subcellular localization data on the three PPI datasets may indicate that subcellular localization data is rich with potential features; 3) To enhance the model's efficiency, we will try to simplify the model architecture to shorten the number of parameters and running time.

## Code Availability

The source code can be downloaded from https://github.com/zzmaoyuqing/EPViT/tree/master.

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